

THE LEDGER FRAMEWORK

Market Data Manifesto

Governing Principles for Market Data — Subordinate to the Architectural Constitution

Version 1.1 — amends the certified 1.0 — July 2026

The record holds observations; every price beyond them is derived from them.

Abstract

The Ledger records observations, not prices. A market data fact enters the record exactly once, as a moveless observation admitted through the one door; every curve, surface, correlation, and calibration is a deterministic function of recorded inputs, of two kinds the document keeps apart. A *projection* computes over the record and rebuilds from it alone; a model's output *re-enters as a recorded observation* (C-14.9, C-14.15). Reading back any recorded value is unconditional; re-deriving a model's output is what needs the model. A value is meaningful only in a *frame* and at a time coordinate — corporate actions are first-class events that change frames; every datum is bound to a model by recorded lineage and validated in price space; and every datum is dispute-ready, settled by replay. This manifesto states the principles that make append-only, event-driven, projected market data the only sensible way to hold it. Each principle is an *article* derived from six of the Constitution's eight commitments — auditability, reproducibility, time travel, correctness, order, simulability — and a seventh it *derives* from the first two, dispute-readiness; each rests on the Constitution clause that carries it. The document is rated against the Constitution, never the other way round; where the two conflict, the Constitution prevails.

1. Market Data in the Ledger's Terms

A market data fact is an **observation**: a recorded statement, attributed to a source, that a named observable — a price, a rate, a fixing, an index level — took a value with respect to a world-moment. The observation is a fact about what a source said, not the truth about what a thing is worth; any “true” price behind it is a modelling posit that lives in a model, never a claim the record makes.

The source of an observation sits outside the trust boundary; the Events Executor captures and records the arrival (C-5.4). It enters the immutable log only as a moveless *observation-recording transaction* — moveless because it records data and moves no balances — admitted through the single Transaction Executor and folded into a *home*, its resting place in the record from which contracts read (C-4.8). (To *fold* the log is to replay it in order, accumulating state; the current state is the initial state plus the fold of all observations.) There is no second door for market data: the one record and the single writer extend to every observation. A kind of market data is declared before any observation of it crosses — its fields, the frame it is delivered in, the market data operators that adjust it across corporate actions (C-9.2), and the grain of the cause-derived identifier — an identifier computed from what caused an arrival, so a redelivery of the same cause collides with it — that decides when two arrivals are the same fact, all registered on the record first, so an arrival of an undeclared kind is quarantined, never guessed (C-4.12).

Vocabulary, mapped once. This manifesto uses the Constitution’s fixed names and coins no synonym. Its prior art, the Market Data Calibration Manifesto, speaks a different dialect, mapped here once and not carried further: an *Oracle* is an external source (no “oracle” primitive, MD-3); an *attestation* is an observation (C-4.8); *event time* is execution time, *knowledge time* splits into monitor and door time (C-2.7); a *posterior* is a derived object (MD-6); a *model configuration* is the recorded, versioned terms of a derived object (C-4.4, C-7.2). The Constitution’s own term *the market data operator* names one thing only — the corporate-action transform of C-9.2 — and is never used for a source; version 1.1 uses that operator as the map between *frames* (MD-13). *Frame* builds on the Constitution’s own “post-event frame” (C-9.2), which this manifesto promotes to a defined coordinate system — the one term it fixes that the Constitution names only informally. No synonym for the operator is introduced.

2. The Articles

Each article derives from one or more of the six commitments and rests on a clause of the Constitution. The identifiers MD-*n* are append-only: never reused, never renumbered, so later documents may cite them.

MD-1. The observation is the atom of market data. A market data fact is an observation, MD-1 and an observation is recorded exactly once. It enters the immutable log as a moveless observation-recording transaction, admitted through the single writer, and folded into a home. “Exactly once” is a mechanism, not a wish: each arrival carries its cause-derived identifier, and an arrival identical under it is *absorbed* as a duplicate, not recorded again (C-2.7, C-12.3, C-14.3). Absorption is not a silent drop: no new value is recorded, but the redelivery’s arrival provenance is retained, so a healthy absorb is distinguishable from a feed gone quiet. Three arrivals that look alike must be told apart, and the identifier’s grain — a registered property of the data kind (§1) — is what tells them: the same print *redelivered* is absorbed; a vendor *correction* is a distinct, later observation that names the value it corrects (MD-10); a genuinely *new* print at the same instant is a second observation, recorded in its own right. A grain too *fine* double-counts a redelivery; a grain too *coarse* absorbs a genuine second print and loses it — and that the registered grain is correct is a trust assumption, named beside TA-ARRIVAL. The double-count is loud, the extra observation on the record to be found; the over-coarse loss is *silent*, and where two distinct prints are identical under every field a feed sends, no grain can separate them. This one residual silent loss the manifesto does not pretend to prevent; it is caught where TA-ARRIVAL’s gaps are — at the perimeter, reconciling arrival counts against recorded observations. Subject to that, nothing about market data is written twice: the value a source stated is its sole canonical copy, computed from wherever it is mentioned, never stored beside it.

(*Auditability, Correctness; C-1.4, C-4.8, C-2.7, C-12.3, C-14.3.*)

MD-2. Capture precedes judgement. The record admits an observation before anything MD-2 decides whether to believe it. A stale, fat-fingered, or off-consensus quote is well-formed: it enters through the normal door as an observation-recording transaction (C-4.8), and judgement is deferred because economic correctness is never gated at admission — it is detected after the fact by recomputation (C-13.2, C-13.3). A quote that is not even processable — a corrupted payload, an unresolvable reference — is still not lost: it is captured with the provenance of its arrival and stands as a visible open item (C-4.12). Either way the Ledger captures, then classifies; it never refuses to record in order to reject. Whether an observation is used, down-weighted, or ignored is a separate and later question, answered by a derived object over the record, never by the door. A “rejected” observation is one a calibration assigned no weight — present, inspectable, reclassifiable later — not one the record never held. Capture-then-classify is a *principle* whose guarantee runs *from arrival*: whether a pipe can hold the open’s firehose is a capacity question

outside this document (TA-ARRIVAL), but the principle forbids using capacity as a licence to drop silently — what the boundary cannot admit is a named, visible residual; what never reaches it is the perimeter’s concern (C-4.12).

(*Auditability, Order, Correctness; C-4.8, C-13.2, C-13.3, C-4.12.*)

MD-3. The source is untrusted; its delivery is a trust assumption. The origin of an observation — an exchange, a broker, a fixing provider, an on-chain feed — is machinery beyond the system’s trust boundary, an external source (C-5.1, C-5.8), and the framework grants no source the standing of truth. That an observation reaches the boundary at all is a trust assumption, reconciled at the perimeter, never a fact the fold may assume (C-4.12). The framework has no privileged “oracle” component: a source is where an observation came from, recorded as provenance, and nothing more.

(*Auditability, Correctness; C-5.1, C-5.4, C-5.8, C-4.12.*)

MD-4. Every observation bears three times, and as-of and as-at are two of them. By the Constitution an event carries an execution time — the world-moment its source asserts it happened, the time a court would enforce, contestable only in the world and corrected only by a later event; a monitor time — when the boundary observed it; and a door time — when the single writer admitted it (C-2.7). The *as-of* coordinate — the world-moment a value concerns — is its execution time; the *as-at* coordinate — when the record came to hold it — is its door time; monitor time is provenance, splitting lateness into the world’s delay and ours. Because both are recorded, the system computes both honest answers at any moment: what was known as of a past date, and what is known now about it (C-12.1). A value carries one coordinate more: the corporate-action frame it is expressed in (MD-13).

(*Time travel, Order; C-2.7, C-12.1.*)

MD-5. Order is meaning, and the authoritative fold is execution-ordered. Observations are events, and their sequence is load-bearing: a derived object depends on which observations preceded it — a surface fitted before a shock is not the surface fitted after it. Meaning lives in execution order. An observation that arrives late, whose execution time precedes observations already folded, takes its place among them; every **projection** downstream refolds to include it (C-2.7), while a **re-entered observation** whose input window it lands in cannot re-fold itself — the ledger runs no model (C-14.9) — and is instead flagged stale for re-derivation (MD-8). Neither is silent: the insertion and every changed or staled value are flagged, and the difference is published as a named explain item — a line in the profit-and-loss explain (C-12.6). The as-known view — what the book showed with only the observations it then held — survives unrewritten beside the corrected one.

(*Order, Time travel; C-2.7, C-12.6, C-14.9.*)

MD-6. Every derived object is a deterministic function of recorded inputs. A curve, a surface, a correlation, a calibration is a function of what the record holds — the observations and any derived objects it consumes, each pinned to its version and *cut* (the as-at boundary fixing which observations were in force; MD-12), the recipe that defines it, and, where the recipe draws on randomness, a recorded seed. Two kinds of derived object must be told apart, and the Constitution names both.

Where the recipe computes *over the record* — a discount factor read off recorded quotes, an adjusted value from the market data operator — the object is a **projection**: it stores nothing and rebuilds from the record alone, by any party, today or at a past date (C-2.2, C-3.2, C-14.11). Where the recipe *runs a model* — a filter, an optimiser, a fitted surface — the ledger does not run it (C-14.9); the model is evaluated outside the fold and its output re-enters through the

one door as a new **observation** (C-14.15). That number is stored like any observation, and two senses of recovering it must not be confused: *reading it back* is always possible from the record alone; *re-deriving* it — showing it follows from its inputs — needs the retained model and the numerical environment it ran in, a governance matter outside this scope. A re-entered observation is read back unconditionally, re-derived only with the model.

Either kind carries, as recorded provenance, a **complete lineage**: the resolved input cut — every observation and derived object it consumed, each pinned to version and cut; the corporate-action events whose frame the operator applied (MD-13); for a model output the model version; and for any datum used in valuation the model it is bound to (MD-15). Lineage reaches through the whole chain to the leaf observations, so what an object depends on — including the corporate actions behind its frame — is always legible. A derivation that *fails* is not an absence, and its treatment follows the split: a **projection** failure — a bootstrap with too few quotes — is legible by recomputation, never stored, and refreshes to success if a late arrival fills the gap (MD-5); a **model-run** failure re-enters as a recorded observation with its diagnostics, itself stale if its inputs later move; a failed *capture* (MD-2) is stored as an input-side arrival, not a derivation. An absent object never passes for one never attempted. Record enough — observations, derived inputs, recipe, seed, lineage — that the object is a deterministic function of the record.

(*Reproducibility, Simulability, Correctness; C-2.2, C-3.2, C-14.9, C-14.11, C-14.15.*)

MD-7. The prior is a term; the posterior is a derived object. A Bayesian calibration is not banned; it is placed. The prior — including any assumed dynamics for how a parameter moves between observations — is a declared, recorded term. The posterior is a derived object of that prior-term, the recorded observations, and the seed (MD-6): a projection where it computes over the record, a re-entered observation where a filter evaluates a model. Either way it is *not* stored state — no “current posterior” updates in place; today’s posterior is tomorrow’s prior names a recomputation, not a mutable variable. The framework mandates no dynamics — a random walk, a martingale, a mean reversion are recipe choices, never framework axioms — and whatever is declared is recorded, so the posterior is reproducible on MD-6’s terms.

(*Correctness, Reproducibility; C-2.8, C-4.11, C-14.15.*)

MD-8. A broken state cannot hide. A recurring pathology of calibrated systems is the stored parameter that has drifted from what the information implies: the system knows something it has not yet written into its numbers. The Ledger defeats it by a mechanism that differs by kind. A **projection** stores nothing — recomputed from the record on every read (C-4.11) — so it cannot lag the record: the drifted-parameter state is unrepresentable, and under a joint recipe an on-demand read is the correct joint conditional expectation, so even two correlated legs cannot hold a divergent pair. A **re-entered observation** does store a number — a fitted surface, or a joint vol-and-correlation fit for a spread, sits on the record because the ledger runs no model (C-14.9). It cannot drift in place: a new fit is a new observation, forward, naming the inputs it consumed (MD-6), never an edit of the old. And when an input it consumed is later corrected or superseded — a wrong price, or a corrected corporate action behind its frame, both pinned lineage inputs (MD-10, MD-13) — the ledger does not silently recompute it; its lineage shows the input moved under it, and the stale fit stands as a flagged open item for re-derivation, so the joint spread surface cannot silently disagree with a corrected leg or a restated split. That visibility is reached by default: a consumer reads a re-entered observation through a projection — “the current fit for X”, selecting the latest non-superseded observation and carrying its staleness flag — never the raw stored value. What the framework forbids everywhere is the *silent* divergence: for a projection it cannot arise; for a re-entered observation it cannot hide.

(*Correctness; C-1.5, C-4.11, C-11.5, C-14.9.*)

MD-9. Constraints live in the derived object; no-arbitrage is declared, not assumed. MD-9

A recorded observation carries no promise of mutual consistency — a stale quote and a fresh one may cross, and two sources may disagree; the record holds both. No-arbitrage, smoothness, and monotonicity are constraints a derived object imposes on its own output, not properties the record must have. A derived object that cannot meet the constraints it declares records that it cannot — a visible diagnostic — rather than serving an inadmissible object as though it were sound. No-arbitrage is not negotiable for the object that claims it, but it is enforced by detection, not admission: an arbitrageable surface served as arbitrage-free is an economic error caught after the fact by recomputation, not prevented at a door (C-13.2, C-13.3), with claim and test both on the record. Which no-arbitrage conditions are correct, and whether a fitted object hedges, are model choice, outside this scope (C-Scope.11). No-arbitrage is the object’s *internal* consistency; its *external* validity — whether it reprices what it was drawn from — is decided in price space (MD-15); both are recorded diagnostics, neither prevented at a door.

(*Auditability, Correctness; C-13.2, C-13.3, C-Scope.11.*)

MD-10. Corrections repair forward; the original is never overwritten. A market data MD-10

fact found wrong is not edited. The discovery is a new recorded observation that names the wrong one, and the discrepancy stands as a visible open item. What happens downstream splits by kind: every **projection** over the corrected fact recomputes automatically on the next read; a **re-entered observation** that consumed the wrong fact cannot re-run itself (C-14.9), so its recorded lineage marks it stale and it stands as a flagged open item until a fresh fit supersedes it forward (MD-8). A bulk retraction — a vendor voiding a whole day — is many corrections at once; the explain may summarise them as one named item rather than flood the book, so the signal is not lost in the flags. Where money already moved on the strength of the wrong fact, it moves back only as an authorised compensating transaction (C-12.4). The observed value stays on the record exactly as observed. Adjustments a later event forces on earlier data — a split rescaling a price, a dividend rescaling a cash-per-share figure — are a *change of frame*, not a correction: computed at each read by the market data operator (C-9.2, C-9.3), declared once per data-and-event pairing and never improvised, obeying the algebra of MD-13. The original is preserved, the adjusted value derived at read, and derived quantities are recomputed from operator-adjusted inputs (C-9.3), never scalar-transported.

(*Auditability, Time travel, Correctness; C-9.3, C-12.4, C-12.6, C-14.9.*)

MD-11. Simulated market data is real market data under a different seed. The state MD-11

of the market data universe is the initial state plus the fold of all recorded observations (C-2.8). A stress scenario, a backtest, a Monte-Carlo path is not a separate system but the same machine folding a generated stream of observations from a recorded seed; production is simply the one path whose observations happened to be real. Because the seed is the single non-record input and it too is recorded, every simulated path replays exactly, on the same terms as one over real observations — but only because every model output on the path is itself captured as a re-entered observation (MD-6); a path that recorded only the seed could not replay.

(*Simulability, Reproducibility; C-2.8, C-10.1.*)

MD-12. Derivation composes. A derived object may consume another derived object — a MD-12

volatility calibration reads a discount curve, a spread model reads two surfaces. When it does, the dependency — which object, at which version and cut — is part of the recorded recipe (MD-6), so the composite is again a deterministic function of the record and the MD-6 split composes: a chain of **projections** reconstructs bit-for-bit from the record; a chain through a **re-entered observation** enters that node as a *leaf* — read back always, re-derived only with its retained model (MD-6). Because each input is fixed at its own cut, never a mutable store,

the “mid-update input” cannot arise (MD-8): a curve feeding a calibration is consistent with it by construction, not scheduling.

(*Reproducibility, Correctness; C-2.2, C-14.11.*)

MD-13. A corporate action is a change of frame; the operators that make it compose. MD-13

A corporate action is first-class — a recorded transaction like any other (C-9.1), never an adjustment applied to market data outside the ledger. It re-coordinates every value recorded before it (C-9.2), so a value is not a bare number but lives in a **frame**: the coordinate system fixed by the corporate-action terms in force *and* the declared adjustment convention (which actions the operator applies, price- or total-return, the rounding rule). The frame a value arrives in is itself an *asserted, recorded fact* of the observation — provenance registered per data kind (§1), never inferred from execution time, since a raw feed delivers unadjusted but a vendor’s back-adjusted series does not. The operator transports from the *declared* delivery frame, so a value already adjusted at source is never adjusted again.

A corporate action carries its own three times. Its **ex-date** is its as-of — the moment the price re-coordinates, the frame boundary — distinct from when its terms became known. Before ex, or if the action is cancelled, the operator is the identity on the price series, so a cancelled-before-ex action never forces an un-adjust; an announcement, or a revised dividend, is not a frame change but new information that refolds projections (MD-5). And the operator is determined only once the terms are *resolved*, not merely announced: where the map turns on a later fixing, proration, or holder election — a scrip reference price, a merger’s blended consideration — it exists from the *resolution* observation, and before that the frame is provisional and legible as such (MD-6), never a silent wrong number.

The map is the Constitution’s **market data operator** (C-9.2), a projection computed at read, the original never overwritten (C-9.3, MD-10). It acts on *leaf* observations and is not in general a proportional scalar — a special cash dividend shifts a forward additively, a listed option’s OCC adjustment re-coordinates strike, multiplier, and deliverable. Derived objects — surfaces, curves, correlations — are never transported by an operator of their own; **derived quantities are recomputed from operator-adjusted inputs** (C-9.3, MD-6, MD-12). Operators compose at *full precision*, the minor-unit rounding (C-4.6) applied *once* at read, so the composite is grouping-independent and two parties reconstruct the identical value; rounding between operators would not.

These operators form a plain algebra over frames. They **compose**: two splits a week apart chain forward into one map, associative because it is function composition. Over an action-free span the operator is the **identity**, and some actions are the identity for a data kind — a proportional dividend figure stands across a split (C-9.2). An **inverse** exists only where the transform preserves information; under rounding most operators lose it and have none, and the framework needs none — the original is preserved (C-9.3), so any frame is reached forward from the execution-time frame, never by inverting. The operator changes a value’s frame and leaves its as-of and as-at (MD-4) unchanged, so it *commutes* with as-of selection under a fixed as-at cut — adjust Tuesday into today’s frame then pick Tuesday, or pick then adjust, gives the same value. The cut must be fixed: an action whose door time falls between two cuts is known only at the later, so the composite operator differs between them. A corrected or late corporate action is a *superseding* event (MD-5, MD-10): every projection over the old frame recomputes, and — the action being a pinned lineage input (MD-6) — every re-entered observation whose lineage reaches it is flagged stale, exactly as a corrected price does. Hence **a quote is meaningful only given a time coordinate and a frame**: any two parties who agree on (as-of, as-at, frame) reconstruct the identical value.

(*Order, Time travel, Reproducibility, Correctness; C-9.1, C-9.2, C-9.3, C-2.3, C-4.6.*)

MD-14. Every datum is dispute-ready. Auditability and reproducibility, pressed to their adversarial conclusion, give a seventh governing principle: a dispute about a recorded value is settled by *replay*, not argument — dispute-readiness is what those two principles already are once an adversary insists, asking nothing new of the record. To contest a mark is to contest a number, and the record answers by exhibiting the whole chain that produced it — the datum, its provenance and attestation, the **cut** it was struck at, the **frame** it was read in (MD-13), and the model it is bound to (MD-15) — and reproducing the valuation bit-for-bit (MD-6), the as-at pinned to the mark’s own cut, so the number that stood then is reproduced, not a post-correction one. Its bound is exact: replay settles whether the value is the faithful, deterministic consequence of the inputs the record held at its stated (as-of, as-at, frame) — the only question a system of record can answer. It does *not* settle a two-sided economic dispute: a counterparty’s mark, from its own surface, model, and frame, replays bit-for-bit too, and which is right is model choice, out of scope (§4). But exhibiting the frame and the bound model *localises* the disagreement — one side applied the split to the strike, the other to the multiplier — which is the value replay does deliver. Its reach is MD-6’s: a value reproduces from recorded prices unconditionally, a model’s number once the model is supplied (C-14.15), never wider.

(*Dispute-readiness (Auditability + Reproducibility); C-2.1, C-2.2, C-2.3, C-Scope.11.*)

MD-15. A datum used in valuation is bound to a model; validity is decided in price space. A market data value on its own is a number; it becomes a fact of the valuation universe only when bound to the model through which it prices something. The binding is not a claim that the model is true — this manifesto asserts no model, and a dynamics such as the martingale, or a latent “true price”, is admissible only as a declared, recorded term, never an axiom. The binding is **recorded lineage**: which model, which version, stands beside the datum (MD-6), exactly as an input cut does.

Why price space: no one argues a printed number is wrong in itself. Disputes are about the valuation of *units* (MD-14), and a unit’s value is defined only in price space, where truth is contractually fixed. So a datum is validated where it is used: it is **valid on its calibration set** when, through its bound model, it reprices the instruments it was drawn from within a declared tolerance — off-set use is model extrapolation, whose validity is model choice, not this round-trip. The round-trip is the acceptance test, and it shows calibration and validation to be one act seen from opposite directions — calibration reads parameters from prices, validation reads prices back. Both are derived objects over recorded inputs and the declared model term (MD-6), so model-binding never undoes the split, and the repricing residual is itself a re-entered observation — a recorded diagnostic (MD-9), stale if its inputs later move (MD-8), never a silent pass. Which model, and which tolerance, are model choice, out of scope (C-Scope.11); that validity is decided in price space is not.

(*Correctness; C-2.4, C-13.2, C-14.15, C-Scope.11.*)

3. The Life of an Observation

One observation can walk the whole picture. Follow AAPL’s official close on a Tuesday, needed by an equity volatility surface.

- **Recorded once, in a frame.** The exchange publishes AAPL’s Tuesday close of 150.20; the Transaction Executor admits a moveless observation-recording transaction. The value is on the record once (a redelivery is absorbed), stamped with execution time (Tue 16:00, as-of), monitor, and door time (Tue 18:30, as-at), in the pre-split frame it was quoted in (MD-1, MD-4, MD-13).

- **A split is a change of frame.** The next week a two-for-one split is recorded as a first-class event. The market data operator carries 150.20 into the post-split frame as 75.10, computed at read; the original is never overwritten, and any later action’s operator chains onto the split’s, forward, in execution order (MD-13, MD-10).
- **Corrected forward, two answers.** On Thursday the exchange restates the close to 150.50 — a correction (a new observation naming the old, MD-10), distinct from the split. The two effects separate: the split alone reframes 150.20 to 75.10; the correction alone lifts 150.20 to 150.50. So *Tuesday as published Tuesday* (cut as-at Tue 18:30, pre-split frame) is 150.20, while *Tuesday as valued now* (cut as-at today, post-split frame) is the corrected close reframed, 75.25 — a quote needs both a time coordinate and a frame.
- **Disputed.** A counterparty contests a collateral mark that used Tuesday’s surface. It is not argued: replay exhibits 150.20, its provenance, cut, frame, and bound model, and reprises the mark bit-for-bit, the as-at pinned to the mark’s cut (MD-14). Replay cannot decide whose surface is right, but it *localises* the disagreement to a differing frame, input, or model.

4. What This Manifesto Does Not Govern

This manifesto governs how market data is recorded, ordered, corrected, and projected. It does not govern **price formation and discovery** (where a price comes from is not a question the record answers); **which model, prior, or no-arbitrage condition is correct** (model choice, whose test is hedging performance); **model governance, validation, and retention** (a value reproduces from recorded prices unconditionally, a model’s number only if the model is kept); **the terms of a corporate action** (the operators transport data through it, MD-13; the terms arrive as recorded events, not decided here); and **settlement** (the record projects committed activity toward it but does not perform it). These lie outside the Ledger boundary and are reconciled against it only at its perimeter (C-Scope.11).

5. Subordination

This manifesto is subordinate to the Constitution and rated against it, never the other way round. Its vocabulary is the Constitution’s, one name per component, no synonym for any fixed term. Where a principle here would conflict with the Constitution, the Constitution prevails and the conflict is parked with the exact amendment text and the exact clause it replaces — never resolved by quiet narrowing. A genuine departure amends this document explicitly first; an article identifier is never reused or renumbered.

Conclusion

Reconciliation of market data has the same root cause as every other: the same fact stored more than once, and the Ledger removes it. An observation is recorded once; every curve, surface, and calibration is a deterministic function of what the record holds — read back always, re-derived only where a model runs; every value carries its times and its frame, is bound to a model, and is settled by replay; corrections repair forward, originals survive. Operate it any other way and you reintroduce the divergence the architecture exists to remove; this way, there is nothing left to reconcile.

Amendment record — Version 1.1 embeds three mandates in the certified 1.0: corporate-action frames and the operator algebra (MD-13); dispute-readiness, a derived seventh principle (MD-14); and model binding with price-space validation (MD-15). MD-4, MD-6, MD-9, MD-10 extended; MD-1–MD-12 keep their numbers.